BDIC1032J Advanced Mathematics (Module 4; Engineering)

SECTION A — GAP-FILLING QUESTIONS

This section is worth a total of **60** marks, with each question worth **5** marks.

1. Find the points on the hyperboloid $x^2 + y^2 - 3z^2 = 8$ where the tangent plane is parallel to the plane $x - 2y + 3z = 6$. Answer for the points: ______________________________

2. The plane $x - y + 4z = 10$ intersects the paraboloid $z = x^2 + y^2$ in an ellipse. Find the points on the ellipse that are nearest to and farthest from the origin.

Answer for the nearest point: ________________ Answer for the farthest point: ________________

3. Find equations of the tangent lines to the plane curve $x = 3t^2 - 2$, $y = 2t^3 - 1$ that pass through the point $(1, 1)$. *Hint*: There are *two* such tangent lines.

Answer for the tangent lines: ____________________; ____________________

4. Find the length of one arch of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ where a is a positive constant and $0 \le \theta \le 2\pi$ is the parameter. Answer for the arclength: ____________________

5. Calculate the iterated integrals $\displaystyle\int_0^1 \left(\int_{\sqrt{y}}^1 \frac{y e^{x^2}}{x^3}\, dx \right) dy =$ ____________________

6. Suppose that $f(x, y) + \displaystyle\iint_D f(x, y)dxdy = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}$ and $D = \{(x, y) \mid x^2 + y^2 \le a^2\}$. Then $f(x, y) =$ ______________________________, $\displaystyle\iint_D f(x, y)dxdy =$ ____________________

7. A lamina occupies the part of the disk $x^2 + y^2 \le a^2$ that lies in the first quadrant. Find the centroid of the lamina. Answer for the centroid is ____________________

8. Find the volume of the solid that lie under the paraboloid $z = x^2 + y^2$ and above the region D in the xy-plane bounded by the line $y = 2x$ and the parabola $y = x^2$.

Answer for the volume of the solid is ____________________

9. Find the work done by the force field $\mathbf{F} = \langle z, x, y \rangle$ in moving a particle from the point $(3, 0, 0)$ to the point $(0, \pi/2, 3)$ along the helix $x = 3\cos t$, $y = t$, $z = 3\sin t$.

Answer for the work: ____________________

10. The vector field $\mathbf{F} = \langle e^y, xe^y + e^z, ye^z \rangle$ is conservative. Find a scalar function f such that $\mathbf{F} = \nabla f$. Use this fact to evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the line segment from $(0, 2, 0)$ to $(4, 0, 3)$.

Answer for the potential function: $f(x, y, z) =$ ____________________

Answer for the line integral: $\int_C \mathbf{F} \cdot d\mathbf{r} =$ ____________________

11. Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle 3x^2 + e^{-(y^2+z^2)},\ -xy,\ xy^2 \sin(x^2 y) \rangle$ and S is the surface of the solid region E bounded by the parabolic cylinder $z = 4 - y^2$ and the planes $z = 0$, $x = 0$ and $x + z = 5$.

Answer for the surface integral: $\iint_S \mathbf{F} \cdot d\mathbf{S} =$ ____________________

12. Find the area of the part of the surface $z = x^2 + 2y$ that lies above the triangle with vertices $(0, 0)$, $(1, 0)$ and $(1, 2)$ in the xy-plane.

Answer for the surface area: ____________________

SECTION B — EXTENDED ANSWER QUESTIONS

This section is worth a total of **40** marks, with each question worth 10 marks.

13. **(10 marks)** First use polar coordinates to evaluate the double integral $\int_0^\infty \int_0^\infty x^2 y^2 e^{-(x^2+y^2)}\, dxdy$ and then use this result to determine the single integrals $\int_0^\infty x^2 e^{-x^2}\, dx$ and $\int_0^\infty e^{-x^2}\, dx$.

14. **(10 marks)** Find the volume of the solid Ω that lies above the cone $z = \sqrt{x^2+y^2}$ and below the sphere $x^2 + y^2 + z^2 - 2z = 0$.

15. **(10 marks)** Use spherical coordinates to evaluate the following integral

$$\int_0^2 \int_0^{\sqrt{4-y^2}} \int_{-\sqrt{4-x^2-y^2}}^{\sqrt{4-x^2-y^2}} y^2 \sqrt{x^2+y^2+z^2} dzdxdy.$$

16. **(10 marks)** Consider the electric field $\mathbf{E}(\mathbf{x}) = \frac{\varepsilon Q}{|\mathbf{x}|^3}\mathbf{x}$ where the electric charge Q is located at the origin and $\mathbf{x} = \langle x, y, z\rangle \in \mathbb{R}^3$, $\mathbf{x} \neq \mathbf{0}$, is the position vector. Show that the electric flux of $\mathbf{E}$ through *any* closed surface S that encloses the origin also satisfies $\iint_S \mathbf{E} \cdot d\mathbf{S} = 4\pi\varepsilon Q$.